

10. Address length of IPV6	1	1	3
(A) 16 bit	(B) 32 bit		
(C) 64 bit	(D) 128 bit		
11. What is the purpose of a default routing in a router?	1	2	3
(A) It directs traffic to all networks in the routing table.	(B) It is used when no other route is found in the routing table		
(C) It specifies the best route to use for network communication.	(D) It updates the routing table automatically.		
12. In a link-state routing protocol, what is the first step a router takes after it detects a change in the network topology?	1	1	3
(A) It updates its routing table.	(B) It waits for other routers to send updates.		
(C) It recalculates the best path using the Bellman-Ford algorithm	(D) It floods a link-state advertisement (LSA) to all other routers.		
13. What is the primary limitation of Hamming code?	1	2	4
(A) It cannot correct errors	(B) It requires too much bandwidth		
(C) It cannot detect double-bit errors	(D) It can only correct single-bit errors		
14. Find the parity bit for the following data units 1001011, 0001100, 1000000, 1110111. Assume odd parity.	1	3	4
(A) 1 0 1 1	(B) 0 1 0 0		
(C) 0 0 1 0	(D) 1 1 0 1		
15. How many frames are in transit at a time in Stop and Wait ARQ?	1	2	4
(A) One frame	(B) Two frames		
(C) Three frames	(D) Four frames		
16. For a sliding window of size (n-1) where n is the sequence number of the frames, there can be a maximum of how many frames sent but yet to be acknowledged?	1	3	4
(A) 0	(B) 1		
(C) n	(D) n-1		
17. What is the primary role of an authoritative DNS server?	1	1	5
(A) To resolve recursive queries	(B) To store DNS records for a specific domain		
(C) To forward queries to other DNS servers	(D) To cache DNS responses		
18. HTTP GET method is used to -----	1	1	5
(A) submit data to the server	(B) request data from the server		
(C) delete data from the server	(D) update existing data on the server		
19. Which of the following protocol provides reliable communication?	1	1	5
(A) UDP	(B) WWW		
(C) TELNET	(D) TCP		
20. Select the well-known port among the following	1	2	5
(A) 65535	(B) 1020		
(C) 8080	(D) 10000		

PART - B (5 × 8 = 40 Marks)

Answer **all** Questions

21. (a) Describe the various network topologies in detail with neat sketch	8	2	1
(OR)			
(b) Compare and contrast twisted pair cables and coaxial cables in terms of their construction, usage and performance.			

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|-----|--|---|---|---|
| 22. | (a) Interpret Network Address Translation (NAT). What are the different types of NAT? Write in brief about their functions.
(OR)
(b) Explain in detail different network devices and its applications. | 8 | 3 | 2 |
| 23. | (a) What are the advantages of Link State Routing over Distance Vector Routing in terms of scalability and convergence speed?
(OR)
(b) Discuss the security features of IPv6. How does it improve over IPv4 in terms of security? | 8 | 4 | 3 |
| 24. | (a) Given the following 4-bit binary data words sent over a communication channel:
1101
1010
0111
0001
Calculate the vertical redundancy check (VRC) for each column of data(Consider Even Parity). How would you append the parity bits to form the final transmitted message?
(OR)
(b) Explain the ARQ protocol used for error control. How does it guarantee reliable data transmission? List out the different types of ARQ. | 8 | 4 | 4 |
| 25. | (a) Differentiate between TCP and UDP protocol
(OR)
(b) What is the role of DNS in the functionality of the internet, and how does it facilitate user access to websites? | 8 | 4 | 5 |

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

Marks BL CO

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|-----|---|----|---|---|
| 26. | Sketch the OSI Layer and state the role of each layer in data communication and Discuss how each layer interacts with the layers above and below it to enable communication in a network. | 15 | 3 | 1 |
| 27. | Imagine you are on a video call where maintaining real-time audio and video flow is essential, even if it means tolerating occasional data loss to avoid delays. Identify the protocol likely used for this video call scenario. Explain why this protocol is more suitable for real-time communication than TCP and describe how it handles packet loss and data transmission to prioritize speed over perfect accuracy. | 15 | 4 | 5 |

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